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SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

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Code of Conduct:

I S. Karthikeyan / 192419048 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CSA07 — COMPUTER NETWORKS

Assessment Tool 2 — Concept Mapping: Answers

CO1: Apply the functions of OSI layers in enabling reliable data transmission across networks

Q1. Identify the missing OSI layers

Given partial map: Application → Presentation → _____ → Transport → Network → _____ → Physical.

Completed Concept Map



Missing Layers Identified

Session Layer — sits between Presentation and Transport.

Data Link Layer — sits between Network and Physical.

Explanation — Why these layers are essential for reliable communication

The Session Layer establishes, manages, and terminates the dialogue (session) between two communicating applications. It provides synchronization checkpoints so that if a session is interrupted, communication can resume from the last checkpoint instead of restarting — this is essential for maintaining continuous, organized communication over long exchanges, such as logins or file transfers.

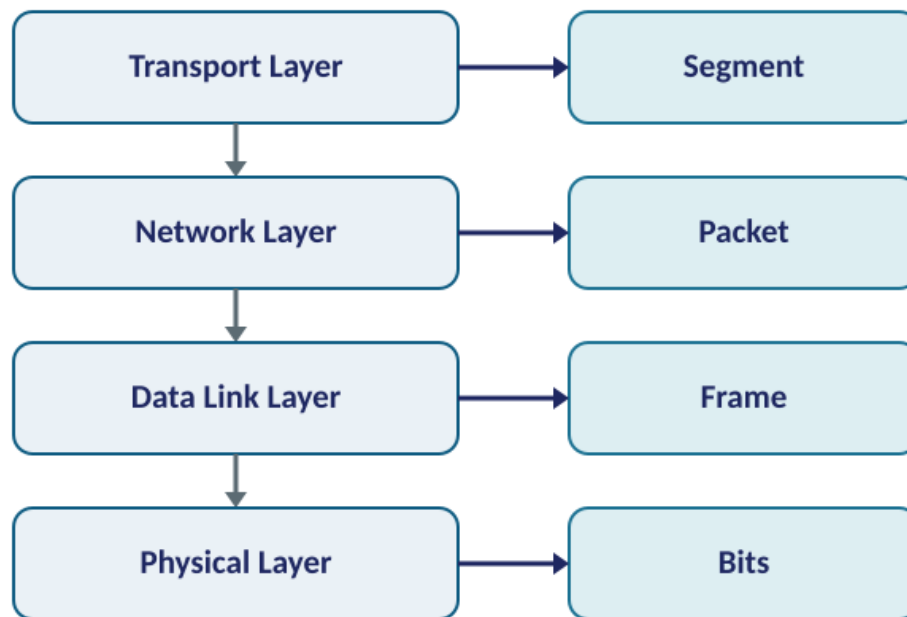
The Data Link Layer takes packets from the Network Layer and frames them with MAC (hardware) addressing and error-detection information (e.g., a CRC checksum) before they are converted to bits. It ensures that data is reliably delivered across a single physical link and that any transmission errors introduced by the medium are detected — and, in many protocols, corrected via retransmission — before the frame is handed to the Physical Layer.

Without the Session Layer, applications would have no structured way to maintain or recover a dialogue; without the Data Link Layer, there would be no reliable, error-checked delivery over the physical medium. Because each OSI layer performs one distinct function, omitting either layer breaks the layered guarantee of end-to-end reliability.

Q2. Data units and their OSI layers

Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers.

Concept Map



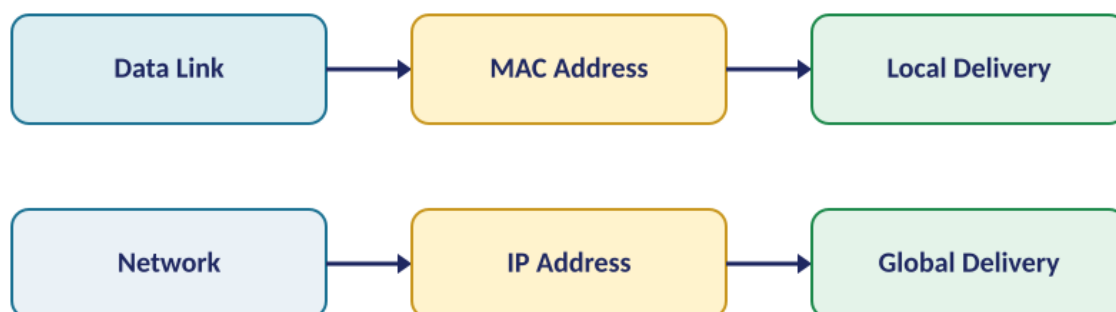
Explanation — How transformation of data units supports transmission

As data moves down the OSI stack, each layer encapsulates the data unit received from the layer above by adding its own header (and, for the Data Link Layer, a trailer too), producing the next data unit in the chain. The Transport Layer wraps application data into a Segment, adding port numbers plus sequence and acknowledgement numbers that enable reliable, ordered delivery. The Network Layer wraps the segment into a Packet, adding source/destination IP addresses for logical addressing and routing across networks. The Data Link Layer wraps the packet into a Frame, adding MAC addressing and error-detection bits for delivery across a single physical link. Finally, the Physical Layer converts the frame into a raw stream of Bits for transmission as electrical, optical, or radio signals.

At the receiving end this process runs in reverse (decapsulation): bits are reassembled into frames, frames into packets, packets into segments, and segments into the original application data. This layered transformation lets each layer solve exactly one problem — addressing, routing, framing, or signalling — making the overall transmission process modular, interoperable between different vendors/technologies, and reliable.

Q3. MAC vs IP addressing — local and global delivery

Given concept map:



Explanation — How these relationships ensure complete data delivery

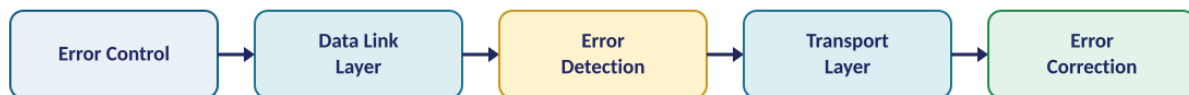
The Data Link Layer uses MAC addresses, which are fixed, hardware-level addresses unique to each network interface. MAC addressing only has meaning within a single local network (LAN) — it lets one device deliver a frame directly to another device it is physically or logically adjacent to on that same segment. This is why it enables "local delivery."

The Network Layer uses IP addresses, which are logical, hierarchical addresses that remain meaningful across different networks. Routers use the destination IP address to make forwarding decisions, allowing a packet to be relayed across many intermediate networks until it reaches a network far from the source. This is why it enables "global delivery."

Together, these two addressing schemes work at different scopes: at every hop along a route, the Network Layer's IP address tells the overall system where the packet must ultimately go, while the Data Link Layer's MAC address tells the current hop exactly which physical device to hand the frame to next. This combination of global logical routing (Network Layer) and local physical delivery (Data Link Layer) is what allows data to travel reliably and completely from a source to a destination that may be anywhere in the world, correctly passing through every intermediate network along the way.

Q4. Error control — detection and correction across layers

Given concept map:



Explanation — How the interaction between these layers improves reliability

The Data Link Layer performs error detection at the level of a single physical link: it attaches a checksum (e.g., a CRC) to every frame, and the receiving node recalculates this checksum to check whether the frame was corrupted during transmission over that specific link. If an error is found, the corrupted frame is typically discarded (or, in some link-layer protocols, a local retransmission is requested), stopping damaged data from propagating further up the stack.

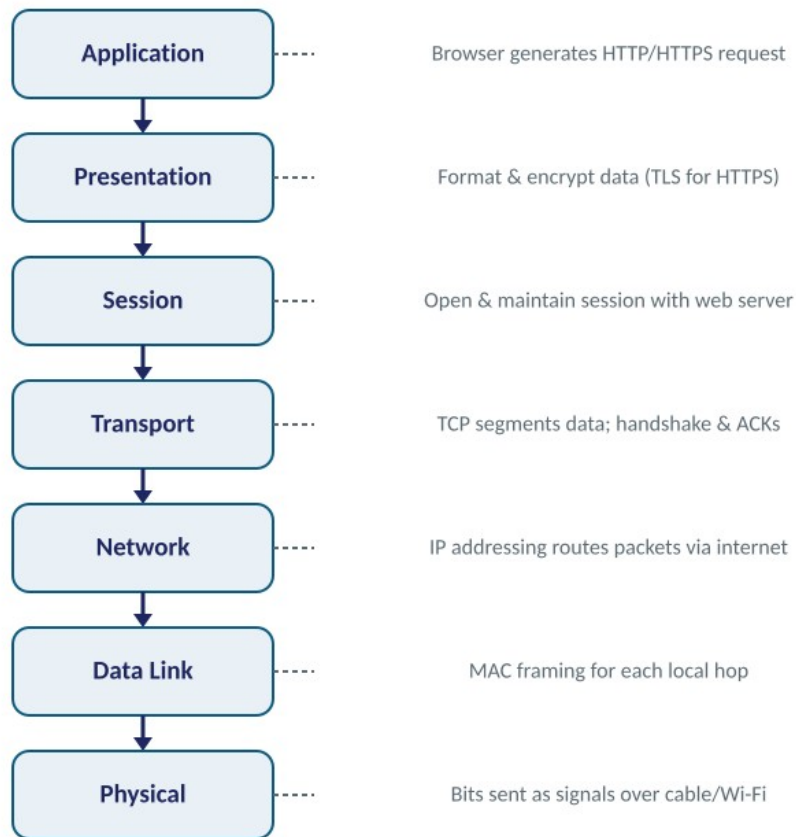
The Transport Layer performs error correction end-to-end, across the entire path from source to destination — which may include many links and routers. Using sequence numbers and acknowledgements (as in TCP), it detects segments that were lost, duplicated, or arrived out of order (including errors that no single Data Link Layer check could catch, since those checks are only local to one hop), and it requests retransmission of exactly the missing or corrupted data.

The interaction between the two layers gives "defense in depth": Data Link error detection catches and discards damaged frames early and cheaply, at the very first link where an error occurs, preventing wasted processing further down the path; Transport Layer error correction then guarantees that the complete, correct byte stream is delivered end-to-end even if isolated errors, drops, or reordering occur somewhere in the network that the link layer alone could not fully resolve. Together, local detection plus end-to-end correction produce a level of reliability that neither layer could achieve on its own.

Q5. Concept map for web browsing

Develop a concept map for web browsing using OSI layers.

Concept Map



Explanation — How failure in one layer affects the entire process

Because every OSI layer depends on the layer beneath it to function, a failure at any single layer breaks the whole chain, even though the fault originates in only one place:

- Physical Layer failure (e.g., an unplugged cable or a dead Wi-Fi signal) — no bits can be transmitted at all, so the page cannot load; every higher layer is inactive.
- Data Link Layer failure (e.g., corrupted frames or a MAC-address problem) — frames cannot be delivered on the local network, so packets never leave the user's own network segment.
- Network Layer failure (e.g., a bad IP configuration or unreachable route) — packets are misrouted or dropped, so the request never reaches the web server even though the local network itself is healthy.
- Transport Layer failure (e.g., a blocked port or failed TCP handshake) — even if packets physically reach the server, no reliable connection can be established, and the browser eventually times out.
- Session Layer failure — the dialogue with the server cannot be maintained across multiple requests, so a logged-in session may unexpectedly drop.
- Presentation Layer failure (e.g., a certificate/encryption mismatch) — the browser shows a "connection not secure" error and refuses to render the page even though the data physically arrived.
- Application Layer failure (e.g., a malformed HTTP request or a server-side error) — the browser displays an error page (such as 404 or 500) even though every layer beneath it worked correctly.

This shows that successful web browsing depends on every layer performing its function correctly — the overall communication is only as reliable as its weakest layer.

Q6. Troubleshooting a network issue using a concept map

Given concept map of a network issue:



Analysis of the problem

The Physical and Data Link layers are both marked as working correctly (✓) — meaning the cable/Wi-Fi connection is up, signals are being received, and local frame delivery (addressing at the MAC level) is functioning normally on the local network segment. The failure (✗) begins at the Network Layer, and because every higher layer depends on the Network Layer to deliver packets, the failure cascades upward, causing the Transport and Application layers to fail as well — even though those layers are not the actual source of the problem.

The most likely root cause sits at the Network Layer: for example, an incorrect IP address/subnet configuration, a missing or wrong default gateway, a broken routing table entry, or a firewall/ACL blocking IP traffic. Since the two layers beneath it are confirmed healthy, the fault cannot be a cabling, NIC, or local-addressing problem — it must be something preventing packets from being correctly routed once they reach the Network Layer.

How the concept map helps in troubleshooting

By showing which layers succeed and which fail in sequence, the concept map immediately isolates the layer where the fault originates instead of leaving the troubleshooter to guess. Because Physical and Data Link are marked good, cabling, switch ports, and MAC-level delivery can be ruled out right away, saving time. Because the chain of failures starts precisely at the Network Layer, the troubleshooter knows to focus directly on IP configuration, gateway reachability, and routing rather than investigating the application code or physical hardware. This mirrors the standard bottom-up, layer-by-layer network troubleshooting method used by network engineers: work upward through the OSI stack, and stop investigating further once the first broken layer is found, since every failure above it is simply a consequence of that one root cause.